

Dr. Amey Pathak

Assistant Professor (Gr-I)

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Google Scholar: <https://scholar.google.co.in/citations?user=do9u7u8AAAAJ&hl=en&oi=ao>;

Citations: 2077; h-index-14; i10-index- 14

(a) Professional Preparation

Indian Institute of Technology Bombay	Mumbai	Water Resources Engineering	Post-doc (2018)
Indian Institute of Technology Bombay Indian Institute of Technology Bombay	Mumbai Mumbai	Civil, (Water Resources Engineering) Natural Resources Engineering (Remote sensing and GIS)	Ph.D. (2017) M.Tech. (2010)
Mahatma Phule Krishi Vidyapeeth	Rahuri	Agricultural Engineering	B.Tech.

(b) Ph.D. Research

Thesis title	Supervisor	Institute	Year
Role of Precipitation Recycling and Moisture Transport during Summer Monsoon in Indian Sub- Continent.	Prof. Subimal Ghosh	I.I.T. Bombay	2017

(c) Appointments

- Assistant Professor (Grade –I) in Land and Water Resources Engineering Specialization, Department of Agricultural and Food Engineering at IIT Kharagpur (19 August 2020-present)
- Assistant Professor (Grade –II) in Land and Water Resources Engineering Specialization, Department of Agricultural and Food Engineering at IIT Kharagpur (02 May 2018 to 19 August 2020)
- Post-Doctoral researcher on project: "*Coupled Human And Natural Systems Environment (CHANSE) for water management under uncertainty in the Indo-Gangetic Plain*" at IIT Bombay (January 2017 to April 2018)
- Lecturer in the Rajiv Gandhi University of Knowledge Technology, Civil Engineering Department, India. (October 2010 to July 2011)

(d) Research Interests

Indian Monsoon Rainfall (Process, Prediction, Projections), Extreme Events, Precipitation Recycling, Geoinformatics applications in Water and Agriculture, Remote Sensing Applications, Agricultural Water Management, AI/ML applications in Agriculture and Climate, Atmospheric Moisture Transport, Climate Change and Climate Modelling.

(e) Publications:

Citations: 2077; h-index-14; i10-index- 14

- Bais, Ranjeet Singh, and A Pathak (2026). **Extreme learning machine based sub-seasonal forecasting of Indian Monsoon rainfall over core monsoon zone**, *Theoretical and Applied Climatology*, Vol. 157, No. 325, pp. 325. Springer Nature; <https://doi.org/10.1007/s00704-026-06274-z>.
- Maurya, A. K., B. R. Behera & A Pathak, (2026). IoT-Enabled YOLO-Based Real-Time Weed Infestation Detection and Alert System for Potato Crop Management. *Potato Research*, <https://doi.org/10.1007/s11540-026-10087-y>
- Kumar, Ranjan, A Pathak, and K Shashikanth (2026). **Climatic and transient controls on Indian monsoon E-P variability using moisture-budget decomposition**. *Environmental Research and Communications*. IOP Science, <https://doi.org/10.1088/2515-7620/ae73ee>.
- R Kumar, and A Pathak (2025) Lagrangian quantification of atmospheric moisture sources for extreme rainfall events over India during the 2023 summer monsoon, *Science of The Total Environment* Volume 1000, 2025, 180389, <https://doi.org/10.1016/j.scitotenv.2025.180389>

- A.K. Maurya, **A. Pathak**, (2025) Leveraging the use of digital agriculture and machine learning for accurate prediction of leaf area index (LAI), *Comput. Electron. Agric.*, 239 (2025), Article 110947
- Bais RS, and **A Pathak (Under Review)**, Analog Sub-Seasonal Forecasting of Indian Monsoon Rainfall over Core Monsoon Zone using Extreme Learning Machine, Submitted to *International Journal of Climatology*
- Shashikanth K., P., Tharun, K., Nagaratna, **A. Pathak**, and H Gupta (**Under Review**) Advanced Machine Learning Techniques for Thunderstorm Prediction for Hyderabad Metropolitan, India", Submitted to *Discover Environment*, Springer Nature Link
- Das K., S. Sarkar, A. Mukherjee, P. Das, **A Pathak**, (2021): Observing tidal and storm generated wave height impact on groundwater levels in a tropical delta (the Sundarbans), *Journal of Hydrology*, Volume 603, Part A, ISSN 0022-1694, <https://doi.org/10.1016/j.jhydrol.2021.126813>. [**Impact factor-4.5**]
- Vittal H., **Pathak A.**, and A. Koppa, (2020) Dual Response of Arabian Sea Cyclones and Strength of Indian Monsoon to Southern Atlantic Ocean, *Climate Dynamics*, <https://link.springer.com/article/10.1007/s00382-020-05577-9>, [**Impact factor-4.486**]
- Das K., S. Sarkar, A. Mukherjee, P. Das, **A Pathak**, (2021): Observing tidal and storm generated wave height impact on groundwater levels in a tropical delta (the Sundarbans), *Journal of Hydrology*, Volume 603, Part A, ISSN 0022-1694, <https://doi.org/10.1016/j.jhydrol.2021.126813>. [**Impact factor-4.5**]
- Chug D., **Pathak A.**, Indu J., Jain S.K., Jain S. K., Dimri A. P., Niyogi D., and S. Ghosh (2020): Observed Evidence for the Steep Rise in the Extreme Flow of Western Himalayan Rivers, *Geophysical Research Letters*, doi: 10.1029/2020GL087815. [**Impact factor-4.58**]
- Sahana, A.S., Pathak, A., Roxy, M.K., S Ghosh, (2018): Understanding the role of moisture transport on the dry bias in Indian monsoon simulations by CFSv2, *Climate Dynamics*, doi:<https://doi.org/10.1007/s00382-018-4154-y>. [**Impact Factor: 4.58**]
- Joseph J., Ghosh S., Pathak A., and A K Sahai, 2018: Hydrologic Impacts of Climate Change: Comparisons between Hydrological Parameter Uncertainty and Climate Model Uncertainty, *Journal of Hydrology*, 566, 1-22. [**Impact Factor: 4.5**]
- Roxy, M. K., Ghosh, S., **Pathak, A.**, Athulya R., Mujumdar, M., Murtugudde, R., Terray, P., Krishnan, R., Rajeevan, M, (2017): A threefold rise in widespread extreme rain events over central India, *Nature Communications*, 8, Article number: 708(2017), doi:10.1038/s41467-017-00744-9 [**Impact factor - 12.124**]
- **Pathak A.**, Ghosh S., Martinez J.A., Dominguez F., and P. Kumar, (2017): Role of Oceanic and Land Moisture Sources and Transport in the Seasonal and Inter-annual variability of Summer Monsoon in India, *Journal of Climate*, 30, 1839-1859, [**Impact factor - 4.85**]
- **Pathak A.**, Ghosh S., R. Murtugudde, and P. Kumar, (2017): Role of Oceanic and Terrestrial Atmospheric Moisture Sources in Intra-Seasonal Variability of Indian Summer Monsoon Rainfall. *Scientific Reports*, Nature Publishing Group, 7, Article number: 12729, doi:10.1038/s41598-017-13115-7 [**Impact factor - 4.259**]
- Paul S., Ghosh S., Oglesby, R., **Pathak A.**, Chandrasekharan, A., and R. Ramshankaran, (2016): Weakening of Indian Summer Monsoon Rainfall due to Changes in Land Use Land Cover, *Scientific Reports*, Nature Publishing Group, 6, Article number: 32177, [**Impact factor - 4.847**]
- Sebastian, D., **A. Pathak**, and S. Ghosh, (2016): Use of Atmospheric Budget to Reduce Uncertainty in Estimated Water Availability over South Asia from Different Reanalyses, *Scientific Reports*, Nature Publishing Group, 6, Article number: 29664, [**Impact factor - 4.847**]
- Vittal, H., S. Ghosh, S. Karmakar, **A. Pathak**, and R. Murtugudde, (2016): Lack of Dependence of Indian Summer Monsoon Rainfall Extremes on Temperature: An Observational Evidence, *Scientific Reports*, Nature Publishing Group, 6, Article number: 31039, [**Impact factor - 4.847**]
- Barik, B., Ghosh, S., Sahana, A. S., **Pathak, A.**, and Sekhar, M., (2016): Water Food Energy Nexus: Changing Scenarios in India during recent Decades, *Hydrology and Earth System Sciences*, 21, 3041-3060, 2017. [**Impact factor - 3.535**]
- **Pathak A.**, Ghosh S., and P. Kumar, (2014): Precipitation Recycling in the Indian Subcontinent during Summer Monsoon. *Journal of Hydrometeorology*, 15, 2050-2066, doi: <http://dx.doi.org/10.1175/JHM-D-13-0172.1>, [**Impact factor - 3.645**]

Conferences:

- Bais RS, and **A Pathak** (2026): Impact of Land Initialization using Target codes for Subseasonal forecasting,

2026/1/29, 106th AMS Annual Meeting, American Meteorological Society, <https://ams.confex.com/ams/106ANNUAL/meetingapp.cgi/Paper/477649>

- Kumar, Ranjan, **Pathak, Amey**, (2025): Tracing Moisture Flow from the Bay of Bengal and Arabian Sea with its Impact on Ganga Basin during Monsoonal Extremes, European Geosciences Union General Assembly 2025 (EGU25), held 27 April-2 May, 2025 in Vienna, Austria. Online at <https://www.egu25.eu/>, id. EGU25-4531, April 2025, DOI: 10.5194/egusphere-egu25-4531
- R Kumar, R., Chouhan, P., & **Pathak, A.** (2024): Identification of Atmospheric Rivers Linked to Extreme Precipitation in the Core Monsoon Zone of India, American Geophysical Union Fall Meeting, AGU24, A07-03, Washington D.C., December 2024.
- Maurya A., and **A. Pathak** (2024): Effects of Water Stress on Spectral Reflectance of Potato Crop Grown in Open Field Conditions, European Geophysical Union 2024, EGU24-4383
- Ranjan Kumar and **A Pathak** (2024): 'Detection and Characterization of Atmospheric Rivers Linked to Extreme Precipitation Events Across the Indian Subcontinent', 2024 International Atmospheric Rivers Conference, Scripps Institution of Oceanography -La Jolla, California June 24-27, 2024, Vol. 5, Issue 67 pp. 73-96), <https://cw3e.ucsd.edu/iarc2024/>.
- Maurya A., and A. Pathak (2023): Spectral Response of Potato Crop under Different Water-stress Conditions, American Geophysical Union 2023. 1386264
- Bais R. S., and **A. Pathak**, (2023): Reliability indices for S2S model for societal application, Seasonal to Subseasonal Prediction Summit, University of Reading, UK 171-171, 2023
- Kumar R., and **A. Pathak**, (2022): Diagnostic Evaluation of Atmospheric Moisture Budget (E-P) during ENSO years and its relation to ISM rainfall variability, American Geophysical Union 2022, H44C-07.
- Bais R.S., and **A. Pathak**, (2023): Reliability indices for S2S model for societal application, <https://research.reading.ac.uk/s2s-summit2023/programme/abstract171/>, WWRP/WCRP S2S Summit 2023, Advancing Sub-seasonal to Seasonal Predictions and their Applications, University of Reading, UK (online).
- Raj A, A Pathak, and K Shashikanth (2024), 'Statistical Downscaling of CMIP6 based Monsoon Rainfall over India and its Assessment', 1552868, American Geophysical Union Fall Meeting, AGU24, A07-03, Washington D.C., December 2024.
- Ranjan Kumar, and Amey Pathak (2024): 'Role of Western Indian Ocean on Indian Monsoon extreme Events during 2023' (Submitted to AGU 2024, Abstract ID: A43Y-03, American Geophysical Union Fall Meeting, AGU23, Washington D.C., December 2024.

Book Chapters:

- Pathak A., Ghosh S. (2019) Land–Atmosphere Interactions in Indian Monsoon at Sub-seasonal to Seasonal Scale. In: Singh S., Dhanya C. (eds) Hydrology in a Changing World. Springer Water. Springer, Cham. https://doi.org/10.1007/978-3-030-02197-9_6.
- Pathak A., S Paul and S Ghosh, 2018: Land-Surface Feedback and Impacts of Land-Use Change to Indian Monsoon Rainfall, Climate Change Signals and Response, Springer, pp 3-20.

(f) Sponsored Project/Consultancy

Principal Investigator

- (i) INFLUENCE OF TERRESTRIAL FEEDBACKS ON RAINFALL AND ASSESSMENT OF DROUGHTS AND EXTREMES UNDER CLIMATE CHANGE
Grant Name: **Core Research Grant** (Earth and Atmospheric Sciences)
Sponsoring Agency: Science and Engineering Research Board (SERB)
- (ii) Water Management plan for the Royal Golf Club of Calcutta (RCGC), Kolkata, West Bengal, India Royal Calcutta Golf Club (RCGC)
Project Type: Consultancy
Sponsoring Agency: Royal Golf Club of Calcutta (RCGC), Kolkata
- (iii) ISIRD-Understanding The Mechanism of Present and Future Variability of Summer Monsoon Rainfall over Ganga Basin

Co-Principal Investigator

- (i) Network Project on Engineering Intervention in Micro-Irrigation System - MIS for Improving Water Productivity Central Institute of Agricultural Engineering (Indian Council of Agricultural Research)
- (ii) PRECISION FARMING DEVELOPMENT CENTRE, The Director of Horticulture Department of Food Processing Industries and Horticulture
- (iii) PFDC Project The Director of Horticulture Department of Food Processing Industries and Horticulture
- (iv) Present day and predicted physico-chemical and climate change influence on solute influx to Northern Bay of Bengal and its impact on biotic system, MoES, NEW DELHI

(g) Industry Collaboration

- (i) Name of Industry: TCS Pune, India (Collaborator(s): Dr. Suryakant Sawant), Type: PG thesis research of Mr. Pradhumn Bansal, Duration: (01-09-2022 to 01-05-2023)
- (ii) Name of Industry: TCS Pune, India (Collaborator(s): Dr. Suryakant Sawant), Type: PG thesis research supervision of Ms. Megha P. Duration: (01-01-2023 to 01-05-2023)
- (iii) Name of Industry: Ruvision Ventures Pvt Ltd, Ahmedabad, India (Collaborator(s): Dr. Hiteshri Shastri), Type: Advisory Panel for Urban centric weather and climate AI foundation model for India.

(h) Computational Skills

- Computational skills: Python, Fortran, MATLAB, R, NCL, CDO, GrADS, ArcGIS, QGIS
- Model: VIC, Dynamic Recycling Model (DRM), Weather Research and Forecasting (WRF)

(f) Thesis Guidance

- PhD- 3 (ongoing); M.Tech.- 28 (+7 ongoing); MS-1

(g) Academic contributions

Teaching	• AG60203: GEO-INFORMATICS FOR LAND & WATER RESOURCES • AG69039: GEO-INFORMATICS FOR LAND & WATER RESOURCES LABORATORY • AG60104: ON-FARM WATER MANAGEMENT • AG69024: ON-FARM WATER MANAGEMENT LABORATORY • AG31003: LAND & WATER RESOURCES ENGINEERING • AG31004: IRRIGATION AND DRAINAGE ENGINEERING • AG39004: IRRIGATION AND DRAINAGE ENGINEERING LABORATORY • CE13003: ENGINEERING DRAWING AND COMPUTER GRAPHICS
Other teaching work	• Contributed in PMKSY, Ministry of Agricultural and Farmer Welfares, Government of India- A report prepared by the AgFE department • Contributed in NPTEL course on Micro Irrigation Engineering (2023-24) • Recorded lectures for an online SWAYAM course: Advances in Agricultural Engineering, Land & Water Resources Monitoring and Management, ARPIT course MHRD, (2019) • Delivered lectures in Advanced Training in Agricultural Engineering (ATE 2018, 2019) for B.E. students of the Institute of Engineering, Tribhuwan University, Nepal, (2019) • Recorded five lectures for a NPTEL course on SWCE, 2018 • AI60002 MACHINE LEARNING FOR GEOSCIENCE (Guest Lectures) at IIT Kharagpur (2019)

(i) Service rendered to other Institutions and Organizations

Invited talks at other institutes

- Schmidt Sciences Workshop on FLOW (Feb 10-11, 2025, IIT Bombay) – hosted by IIT Bombay, and Schmidt Sciences, USA.
- *Atmospheric Moisture Transport During Monsoon and Its Significance at Twilight of the Gods: Linear and Nonlinear Prospects of Climate Change* Symposium, PIK, Potsdam, 21 June 2024 (Online) [Program](#).
- International Climate Research Conclave 2023 (May 26-27, 2023, IIT Bombay) – hosted by IIT Bombay, and MoES & DST, Govt. of India.

- Invited Talk (2024, VFSTR University, Andhra Pradesh) – Extreme precipitation changes and implications in the Indian Subcontinent.
- Invited Lecture (Dec 13, 2023, IIT Kharagpur, AARDO Training Program) – Challenges in Hydroclimate Systems.
- Invited Lecture (Mar 19-24, 2024, IIT Kharagpur, MSME Advanced E-SDP) – Geoinformatics for Crop Health Monitoring.
- Guest Lecture (Mar 1-2, 2024, IIT Kharagpur, MSME Advanced E-SDP) – Geoinformatics in Agriculture, Python & AIML Applications in Agriculture.
- Invited Talk (2024, NCPAH Training Program, Ministry of Agriculture, Govt. of India) – Geo-Informatics for Soil & Crop Health Monitoring.
- Seminar (2020, IIT Indore) – Extreme Precipitation in the Indian Subcontinent: Present and Future (Hydrometeorological & Extreme Events Disaster Risk Management Training).
- Lecture (2019, VJTI Mumbai, TEQIP III Faculty Development Programme) – Understanding Natural Variability of Monsoon Rainfall: Challenges in Hydro-Climatology.
- Lecture (2018, M.S. Patel Dept. of Civil Engineering, Gujarat) – Understanding Natural Variability of Indian Summer Monsoon Rainfall (Climate Change Capacity Building Program).
- Course Development Activity for Post Graduate Diploma in Climate Smart Agriculture and Water Management at Mahatma Phule Krishi Vidyapeeth, Rahuri (2020)

(j) Organized Short Term Course/Training Program

- Short Term Course** (Organised through CEP) **Advanced Entrepreneurship and Skill Development Programme Advanced E-SDP on Agro-informatics (Ministry of Micro, Small and Medium Enterprises)**, Office of the Development Commissioner Micro Small and Medium Enterprises Govt of India Nirman Bhawan 7th Floor Maulana Azad Road New Delhi -110108), Duration: (29-02-2024 to 05-03-2024).
- Training Program:** Assisted in Organizing Customized Training Program on “Precision Agriculture Technologies” from 24th to 30th January 2025 at PFDC IIT Kharagpur- as a Co-PI of PDC project.
- Training Program:** Co-Organized Organizing Customized Training Program on “Precision Agriculture Technologies” from 25th February to 3rd March 2025 at PFDC IIT Kharagpur- as a Co-PI of PDC project.

(k) Contribution to Administrative (non-teaching) work

- **Departmental ERP Representative, In-Charge**, Agricultural and Food Engineering (01 SEP 2020 to present)
- **Departmental Examination, In-Charge**, Agricultural and Food Engineering (01 JUN 2024 to present)
- **Faculty Advisor/Course Coordinator**, LWRE Specialization, Agricultural and Food Engineering (01 SEP 2020 to present)- involved in various academic work prescribed by the HoD, AgFE
- **Farm Management Committee**, Agricultural and Food Engineering (01 JUN 2024 to present)
- **Member, Departmental Academic Committee (PGS & R)**, AgFE department (June 2024 to present)
- **PG Tabulator**, Agricultural and Food Engineering (01 SEP 2020 to present)
- Departmental Maintenance committee-member (01 SEP 2020 to 31 AUG 2023)
- Department Asset Management committee-member (2022-23)
- Departmental Space committee-member (2022-23)
- **Senate member**- Institute Assistant Professor representative
- **Member of Institute’s UG Induction 2024 team.**
- **Member of Institute’s Special Hall Vigilance team 2024.**
- ABIF Selection and Monitoring Committee (ASMC) member
- Member of **new UG curriculum Committee**- Contributed to development of new UG curriculum-department committee member
- Member of **new PG curriculum Committee**- Contributed to development of new PG curriculum-department committee member
- **Prof- In charge** of Field and stall demonstration committee during Krishi Mela 2024 and 2025

- Service in conducting PhD written and Interviews, M.Tech. viva, and various activities as a faculty advisor

(I) Professional and Synergistic Activities, and Awards

- Subject Expert Group (SEG) member of Water Resource management under Unnat Bharat Abhiyan (UBA)- a flagship program of Ministry of Education, Govt. of India (2024)
- PMRF Reviewer for Civil and Earth Science, and Interdisciplinary Sciences (2022-24), (Coordinating Institute: IISc Bangalore, and IIT Bombay)
- Question Paper Setter and External Expert for Summative Assessment Scrutiny, in Vignan Foundation for Science Technology and Research University, Vadlamudi, A.P., India (2024-25).
- Advisory Member during ICCWE-2022, IIT Roorkee
- Selection committee member in C-DAC Pune recruitment (2021)
- IIT Kharagpur *senate member*- Institute Assistant Professor Representative
- Indo-US SERB postdoctoral fellowship (2017-18)
- Served as reviewer for Journal of Climate, Earth System Dynamics, Journal of Earth System Science, Environmental Research Letter, ISH, Computers in Agriculture etc..
- Member of the Indian Society for Hydraulics, American Geophysical Union.
- District level weekly forecast of 2017 heat wave based on the Extended Range Forecast for W.B.
- All India Rank 48 in Graduate Aptitude Test in Engineering (GATE-2008)
- Internship at Regional Remote Sensing Service Centre, ISRO, Nagpur (2007)